

PRACTICAL 19 -

Objective - The name roll number and marks of 10 students are stored in an array named marks. Write a program to find the Name of first, second and third rank holder.

CODE -

```
1  #include <stdio.h>
2  #include <string.h>
3
4  // Structure to store student information
5  struct Student {
6      char name[50];
7      int rollNumber;
8      int marks;
9  };
10
11 // Function to find the index of the student with the highest marks
12 int findHighestMarksIndex(struct Student arr[], int size, int
    excludeIndex1, int excludeIndex2) {
13     int highestMarks = -1;
14     int highestIndex = -1;
15     for (int i = 0; i < size; i++) {
16         if (i != excludeIndex1 && i != excludeIndex2 && arr[i].marks >
            highestMarks) {
17             highestMarks = arr[i].marks;
18             highestIndex = i;
19         }
20     }
21     return highestIndex;
22 }
23
24 int main() {
25     // Array to store information of 10 students
26     struct Student marks[10] = {
27         {"Alice", 101, 95},
28         {"Bob", 102, 88},
29         {"Charlie", 103, 92},
30         {"David", 104, 78},
31         {"Eve", 105, 98},
32         {"Frank", 106, 85},
33         {"Grace", 107, 90},
34         {"Henry", 108, 75},
35         {"Ivy", 109, 96},
36         {"Jack", 110, 89}
37     };
38
39     // Find the index of the first rank holder
```

```
40     int firstRankIndex = findHighestMarksIndex(marks, 10, -1, -1);
41
42     // Find the index of the second rank holder, excluding the first
43     // rank holder
44     int secondRankIndex = findHighestMarksIndex(marks, 10,
45         firstRankIndex, -1);
46
47     // Find the index of the third rank holder, excluding the first and
48     // second rank holders
49     int thirdRankIndex = findHighestMarksIndex(marks, 10,
50         firstRankIndex, secondRankIndex);
51
52     // Print the names of the top three rank holders
53     printf("First Rank Holder: %s (Marks: %d)\n", marks[firstRankIndex]
54         .name, marks[firstRankIndex].marks);
55     printf("Second Rank Holder: %s (Marks: %d)\n", marks[
56         secondRankIndex].name, marks[secondRankIndex].marks);
57     printf("Third Rank Holder: %s (Marks: %d)\n", marks[thirdRankIndex]
58         .name, marks[thirdRankIndex].marks);
59
60     return 0;
61 }
```

OUTPUT -

```
First Rank Holder: Eve (Marks: 98)
Second Rank Holder: Ivy (Marks: 96)
Third Rank Holder: Alice (Marks: 95)
```